

## Question B2

This question carries 20 marks in total.

(a) Given the potential field

$$\phi = x^2z^3 + e^{xy}$$

and the point  $P$  with coordinates  $(1, 0, 1)$ , find

(i) [4 MARKS]  $\nabla\phi$  at  $P$ ;

(ii) [4 MARKS] the rate of change of  $\phi$  in the direction  $\hat{v}$  where  $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ .

(b) (i) [4 MARKS] Use Cartesian coordinates to verify that an arbitrary (differentiable) scalar field  $\phi$  satisfies the vector identity

$$\nabla \times \nabla\phi = \mathbf{0}.$$

(ii) [4 MARKS] The vector field  $\mathbf{F}$  is given by

$$\mathbf{F} = 2xy\mathbf{i} + x^2\mathbf{j} + 2z\mathbf{k}.$$

Calculate  $\nabla \times \mathbf{F}$ . What can you deduce about the vector field  $\mathbf{F}$ ?

(iii) [4 MARKS] Evaluate the line integral of the vector field  $\mathbf{F}$ , given above, between the points  $(1, 1, 2)$  and  $(2, 1, 0)$ . Why does the answer not depend on the path taken?

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## Solutions

**Answer (a)(i):**

Given  $\phi(x, y, z) = x^2z^3 + e^{xy}$ .

The gradient is  $\nabla\phi = \left\langle \frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z} \right\rangle$ .

Calculate the partial derivatives:

$$\frac{\partial\phi}{\partial x} = 2xz^3 + ye^{xy}$$

$$\frac{\partial\phi}{\partial y} = xe^{xy}$$

$$\frac{\partial\phi}{\partial z} = 3x^2z^2$$

Evaluate these at the point  $P(1, 0, 1)$ :

$$\left. \frac{\partial\phi}{\partial x} \right|_P = 2(1)(1)^3 + (0)e^{(1)(0)} = 2$$

$$\left. \frac{\partial \phi}{\partial y} \right|_P = (1)e^{(1)(0)} = 1$$

$$\left. \frac{\partial \phi}{\partial z} \right|_P = 3(1)^2(1)^2 = 3$$

Therefore,  $\nabla \phi$  at  $P$  is:

$$\nabla \phi|_P = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$$

**Answer (a)(ii):**

The rate of change of  $\phi$  in the direction of  $\mathbf{v}$  is given by the directional derivative  $D_{\hat{\mathbf{v}}} \phi = \nabla \phi \cdot \hat{\mathbf{v}}$ .

First, find the unit vector  $\hat{\mathbf{v}}$  in the direction of  $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ :

$$|\mathbf{v}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$$

Now, compute the dot product of  $\nabla \phi|_P$  and  $\hat{\mathbf{v}}$ :

$$\begin{aligned} D_{\hat{\mathbf{v}}} \phi|_P &= (2\mathbf{i} + \mathbf{j} + 3\mathbf{k}) \cdot \left( \frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k} \right) \\ &= (2) \left( \frac{2}{3} \right) + (1) \left( \frac{1}{3} \right) + (3) \left( \frac{2}{3} \right) = \frac{4}{3} + \frac{1}{3} + \frac{6}{3} = \frac{11}{3} \end{aligned}$$

The rate of change is  $\frac{11}{3}$ .

**Answer (b)(i):**

Let  $\phi(x, y, z)$  be an arbitrary differentiable scalar field.

The gradient of  $\phi$  is  $\nabla \phi = \frac{\partial \phi}{\partial x}\mathbf{i} + \frac{\partial \phi}{\partial y}\mathbf{j} + \frac{\partial \phi}{\partial z}\mathbf{k}$ .

The curl of the gradient,  $\nabla \times \nabla \phi$ , is calculated using the determinant:

$$\nabla \times \nabla \phi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

Expanding the determinant:

$$\begin{aligned} &= \mathbf{i} \left( \frac{\partial}{\partial y} \left( \frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left( \frac{\partial \phi}{\partial y} \right) \right) - \mathbf{j} \left( \frac{\partial}{\partial x} \left( \frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left( \frac{\partial \phi}{\partial x} \right) \right) + \mathbf{k} \left( \frac{\partial}{\partial x} \left( \frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left( \frac{\partial \phi}{\partial x} \right) \right) \\ &= \mathbf{i} \left( \frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) - \mathbf{j} \left( \frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x} \right) + \mathbf{k} \left( \frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right) \end{aligned}$$

By Clairaut's Theorem on equality of mixed partial derivatives (assuming they are continuous), the terms inside each parenthesis cancel out to zero:

$$= \mathbf{i}(0) - \mathbf{j}(0) + \mathbf{k}(0) = \mathbf{0}$$

Thus, the identity is verified.

#### Answer (b)(ii):

Given  $\mathbf{F} = 2xy\mathbf{i} + x^2\mathbf{j} + 2z\mathbf{k}$ .

Calculate the curl of  $\mathbf{F}$ :

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 & 2z \end{vmatrix} \\ &= \mathbf{i} \left( \frac{\partial}{\partial y}(2z) - \frac{\partial}{\partial z}(x^2) \right) - \mathbf{j} \left( \frac{\partial}{\partial x}(2z) - \frac{\partial}{\partial z}(2xy) \right) + \mathbf{k} \left( \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(2xy) \right) \\ &= \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(2x - 2x) = \mathbf{0}\end{aligned}$$

Since  $\nabla \times \mathbf{F} = \mathbf{0}$  everywhere, we can deduce that  $\mathbf{F}$  is a **conservative (or irrotational) vector field**. This implies that its line integrals are path-independent, and there exists a scalar potential function  $f$  such that  $\mathbf{F} = \nabla f$ .

#### Answer (b)(iii):

Because  $\mathbf{F}$  is a conservative vector field, the line integral between the two points is independent of the path taken (as established by the Fundamental Theorem for Line Integrals). The answer depends only on the values of the potential function at the endpoints.

Let's find the potential function  $f(x, y, z)$  such that  $\nabla f = \mathbf{F}$ :

$$1) \frac{\partial f}{\partial x} = 2xy \implies f(x, y, z) = \int 2xy \, dx = x^2y + g(y, z)$$

$$2) \frac{\partial f}{\partial y} = x^2 \implies \frac{\partial}{\partial y}(x^2y + g(y, z)) = x^2 + \frac{\partial g}{\partial y} = x^2 \implies \frac{\partial g}{\partial y} = 0 \implies g(y, z) = h(z)$$

$$\text{So } f(x, y, z) = x^2y + h(z).$$

$$3) \frac{\partial f}{\partial z} = 2z \implies h'(z) = 2z \implies h(z) = z^2 + C.$$

Thus,  $f(x, y, z) = x^2y + z^2$  (setting  $C = 0$ ).

Now evaluate the line integral using the endpoints  $A(1, 1, 2)$  and  $B(2, 1, 0)$ :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(2, 1, 0) - f(1, 1, 2)$$

$$f(2, 1, 0) = (2)^2(1) + (0)^2 = 4$$

$$f(1, 1, 2) = (1)^2(1) + (2)^2 = 1 + 4 = 5$$


$$\int_C \mathbf{F} \cdot d\mathbf{r} = 4 - 5 = -1$$